Motor selection

1 Data and Formulas

- Maximum robot mass: $M = 1.3$ kg
- Gravitational acceleration: $g = 9.81$ m/s^2
- Number of motors: 2 identical motors
- Specifications requirement: Nominal speed $v \in [2.0\ ;\ 6.0]$ km/h (4 ± 2 km/h).
- Fall prevention criterion: Minimum acceleration $a_{\max} > 9.81$ m/s^2 ($1.0\,g$).

Formulas

1. **Angular speed:** $\omega = N \times \dfrac{2\pi}{60}$ (rad/s)
2. **Max linear speed:** $v_{\max} = \omega \cdot R$ (m/s) $= 3.6 \times (\omega \cdot R)$ (km/h)
3. **Total ground thrust (2 motors):** $F = \dfrac{2 \cdot \tau_{\text{stall}}}{R}$ (N)
4. **Max acceleration:** $a_{\max} = \dfrac{F}{M}$ (m/s^2) and Margin in $g = \dfrac{a_{\max}}{9.81}$

Characteristics of the 3 Motors

- Motor 1 (FIT0520): $N = 300$ RPM ($\omega = 31.42$ rad/s), $\tau = 3.6$ kg $\cdot$ cm $= 0.353$ N $\cdot$ m ($2\tau = 0.706$ N $\cdot$ m).
- Motor 2 (FIT0521): $N = 210$ RPM ($\omega = 21.99$ rad/s), $\tau = 10.0$ kg $\cdot$ cm $= 0.981$ N $\cdot$ m ($2\tau = 1.962$ N $\cdot$ m).
- Motor 3 (FIT0522): $N = 100$ RPM ($\omega = 10.47$ rad/s), $\tau = 6.5$ kg $\cdot$ cm $= 0.637$ N $\cdot$ m ($2\tau = 1.275$ N $\cdot$ m).

2 Maximum Speed Calculations ($v_{\max}$)

Motor	$R = 3.5$ **cm**	$R = 4.0$ **cm**	$R = 5.0$ **cm**	$R = 6.0$ **cm**
FIT0520 (300 RPM)	3.96 km/h	4.52 km/h	5.66 km/h	6.79 km/h *(Too fast)*
FIT0521 (210 RPM)	2.77 km/h	3.17 km/h	3.96 km/h	4.75 km/h
FIT0522 (100 RPM)	1.32 km/h	1.51 km/h	1.88 km/h	2.26 km/h

Table 1: Maximum linear speed as a function of wheel radius (allowed range: 2 to 6 km/h).

Speed Conclusion:

- **FIT0522 (100 RPM):** Eliminated. Far too slow for wheel radii from 3.5 to 5 cm (< 2.0 km/h, non-compliant with specifications).
- **FIT0520 (300 RPM):** Compliant up to 5 cm, but exceeds the maximum speed limit at 6 cm (6.79 km/h > 6.0 km/h).
- **FIT0521 (210 RPM):** Compliant. Strictly within the required [2.0 ; 6.0] km/h range for all evaluated radius.

3 Maximum Acceleration Calculations ($a_{\max}$ for $M = 1.3$ kg)

Total thrust is $F = \dfrac{2 \cdot \tau}{R}$ and acceleration is $a_{\max} = \dfrac{F}{1.3}$.

Motor	$R = 3.5$ **cm**	$R = 4.0$ **cm**	$R = 5.0$ **cm**	$R = 6.0$ **cm**
FIT0520 (3.6 kg·cm)	15.5 m/s^2 (1.58 g)	13.6 m/s^2 (1.38 g)	10.9 m/s^2 (1.11 g)	9.0 m/s^2 (0.92 g) *(Fall)*
FIT0521 (10.0 kg·cm)	43.1 m/s^2 (4.40 g)	37.7 m/s^2 (3.85 g)	30.2 m/s^2 (3.08 g)	25.2 m/s^2 (2.56 g)
FIT0522 (6.5 kg·cm)	28.0 m/s^2 (2.86 g)	24.5 m/s^2 (2.50 g)	19.6 m/s^2 (2.00 g)	16.4 m/s^2 (1.67 g)

Table 2: Maximum available acceleration to prevent tipping for a 1.3 kg robot.

Acceleration Conclusion:

- To stabilize the inverted pendulum, the acceleration must strictly exceed gravity ($a > 1.0\,g (= 9.81$ m/s^2)).
- **FIT0520 (3.6 kg·cm):** As soon as the radius reaches 6 cm, acceleration drops to $0.92\,g$ ($< 1.0\,g$): the robot cannot recover and tips over. Even at 4 cm ($1.38\,g$), the margin is very narrow.
- **FIT0521 (10.0 kg·cm):** Delivers between $2.56\,g$ and $4.40\,g$ of acceleration, guaranteeing that the robot will never stall or tip over.

4 Selected Choice

1. **FIT0522 (100 RPM; 6.5 kg·cm):** Rejected because it is far too slow.
2. **FIT0520 (300 RPM; 3.6 kg·cm):** Rejected due to insufficient torque and risk of balancing failure at 1.3 kg.
3. **FIT0521 (210 RPM; 10.0 kg·cm):** Selected. The only motor that simultaneously satisfies the speed requirement and provides ample acceleration margin across all radii.
4. **Optimal Wheel Radius:** $R = 3.5$ cm to 4.5 cm (diameter of 70 to 90 mm), providing $\approx$ 3.2 km/h and nearly $4\,g$ of acceleration.